Hypercalcemia Mediated by Vitamin D: Etiology, Diagnosis, and Therapeutic Approaches

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Abstract

In the medical profession, hypercalcemia is known as one of the most prominent conditions that is characterized by an elevated calcium concentration in the blood. There are many factors that contribute to this elevation, particularly an increase in the intake of vitamin D, as such, in this paper, we acknowledge the risks of hypercalcemia being mediated by vitamin D while also exploring the therapeutic applications of vitamin D in managing he condition. This approach is a nuanced approach in therapy and treatment plans as it considers the interplay of vitamin D and calcium metabolism, thereby ensuring that the potential benefits of vitamin D is harnessed.

Keywords : Vitamin D, Calcium, Hypercalcemia, Parathyroid hormone, Canser, Heart disease

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List of abbreviations

NO	abbreviation	Full Term
1	VDD	Vitamin D Deficiency
2	PTH	Parathyroid Hormone
3	TRPV6	Transient Receptor Potential Vanilloid 6
4	PMCA1b	Plasma Membrane Calcium ATPase 1b
5	HHM	Humoral Hypercalcemia of Malignancy
6	CX	Calbindin-D (unspecified form)
7	1,25(OH)2D	1,25-dihydroxyvitamin D (Calcitriol)
8	25(OH)D	25-hydroxyvitamin D
9	CT	Computed Tomography
10	MRI	Magnetic Resonance Imaging
11	PTHrP	Parathyroid Hormone-related Protein
12	RANKL	Receptor Activator of Nuclear Factor Kappa-B Ligand
13	ATP	Adenosine Triphosphate
14	Ca	Calcium
15	Mg/dL	Milligrams per Deciliter
16	pg/mL	Picograms per Milliliter

Chapter One

Introduction

1.1 Background on Hypercalcemia and Vitamin D

An overview of hypercalcemia, a condition caused by elevated levels of vitamin D, shows that current research focuses on all the detrimental effects of the imbalance as it is known to disrupt the homeostasis of calcium and it leads to many health complications. However, if we consider the varied nature of vitamin D and its role within the human body when it comes to calcium metabolism and its potential benefits to bone health, such an approach where to be adapted by the medical and scientific community, it could lead to a more beneficial, informed and nuanced treatment for the patients who do suffer from hypercalcemia caused by elevated vitamin D levels.

Hypercalcemia is characterized by elevated calcium levels inside the bloodstream and is a condition that requires careful monitoring. When calcium levels are disrupted [1], patients experience several signs and complications that influence structures inside the body, and create cardiovascular issues along with high blood pressure and arrhythmias [2]. The role of vitamin D in this context is typically critical for keeping healthy bones and regulating calcium levels within the blood, disrupting homeostasis. It aids the reabsorption of calcium from the gastrointestinal tract and the renal tubules of the kidneys. However, when vitamin D is in excess, it causes an elevated absorption of calcium. Excessive vitamin D is a result of either over-supplementation or clinical conditions that cause mass internal manufacturing of it [3]. As calcium plays a crucial function in many cellular procedures and structural functions inside the frame, maintenance of its levels is vital for general health [4]. This interaction demonstrates the sensitive relationships between vitamin D levels, calcium levels and hypercalcemia.

1.2 Overview of Vitamin D Metabolism and its role in Calcium Regulation

When delving deeper into the physiological role of calcium and vitamin D, it's critical to extrapolate how those elements are intricately linked to various other bodily capabilities and how it can have a large impact on usual fitness. To address and understand hypercalcemia, especially that which is mediated by vitamin D, it is important to appreciate the sensitive interaction among these two key players in calcium’s [5].

On the other hand, vitamin D, acts similarly to a hormone inside the body[6]. It is synthesized in the skin upon exposure to sunlight and due to modern medical and pharmacological technology [7]. it can be acquired from meals and dietary supplements. Vitamin D's number one role is to ensure the maintenance and therefore, homeostasis of the everyday blood ranges of calcium and phosphorus, which in turn serves in aiding to shape and keep robust bones. It helps the absorption of calcium and phosphorus from the intestine, a procedure vital for bone mineralization [8].

The body regulates calcium and vitamin D levels via a complicated feedback loop. The kidneys play a pivotal position in this system, converting vitamin D into its energetic shape, which is vital for calcium absorption. When calcium degrees within the blood drop, the parathyroid glands launch parathyroid hormone (PTH). This hormone serves to indicate to the kidney to convert extra vitamin D into its energetic shape, resulting in an increase of calcium absorption in the intestines and decreasing its excretion within the urine. Additionally, PTH indicators the bones to release calcium into the bloodstream. This interplay ensures strong surroundings inside the body, keeping the sensitive balance of calcium and vitamin D within normal ranges [9].

Disruptions to this balance, such as elevated levels of vitamin D, can result in hypercalcemia - an excess of calcium in the blood. This condition underscores the need for a harmonious stability between those critical factors within the frame. The ability of this complicated interaction is critical for diagnosing and treating situations related to calcium and vitamin D imbalances, such as hypercalcemia. Effective management of such situations calls for a complete approach that considers the difficult relationships between those important components of human health.

1.3 Problem Statement Hypercalcemia Mediated by Vitamin D

It is our understanding that the recent landscape of research has focused highly on all the negative consequences and outcomes of high concentrations of vitamin D, but as we know, vitamin D also plays a crucial role in the human body, such as its role in the metabolism of calcium and its potentially beneficial additions to bone health. As such,

1.4 Objective of the Review Paper

This literature review paper aims to conduct a critical analysis of the current body of literature about hypercalcemia, particularly when induced by high levels of vitamin D. It is through a detailed examination of the current research that we aim to identify the potential harms while also exploring the idea alluded to in the earlier section, which is to approach it with a more nuanced and tentative view. As such, the aim of this study is to critically evaluate and investigate the current body of literature of hypercalcemia because of excessive vitamin D levels followed by outlining and advocating for a more nuanced therapeutic approach when mitigating the condition and its symptoms.

Chapter Two

Literature Review

2.1 Etiology of Hypercalcemia Mediated by Vitamin D

As mentioned earlier, disruptions to the precariously sensitive balance and interplay of vitamin D and calcium can result in hypercalcemia. It is crucial that the medical and scientific community understand the spectrum of potential symptoms and manifestations of the medical condition. Naturally, as is the case with many complicated ailments or conditions, these symptoms can vary in the severity, in the level of debilitation and in how it affects the patient's fitness levels. The severity levels are proportional to the duration of the calcium elevation but also to the degree of elevation of calcium in the bloodstream. Not to mention, the systems that hypercalcemia affects can vary.

There are many reasons for why vitamin D elevations and excessiveness of it can cause hypercalcemia, particularly the excessive intake of vitamin D supplements [10]. And the reasons like immoderate intake of vitamin D can purpose hypercalcemia and hypercalciuria. This takes place when large amounts of 25-hydroxy vitamin D are formed and bind to the vitamin D receptor, contributing to hypercalcemia. moreover, in conditions like sarcoidosis, tuberculosis, and lymphomas, hypercalcemia can result from the activity of ectopic 25-hydroxy vitamin D-1-hydroxylase expressed in macrophages or tumor cells, leading to excessive production of lively vitamin D. furthermore, mutations in the 1,25-dihydroxy vitamin D-24-hydroxylase cytochrome P450 gene can impair the degradation of energetic nutrition D, resulting in accelerated serum calcium tiers [11]. However, there are many other causes for it. It is essential for medical professionals to understand the varying and complex etiological factors that also contribute to hypercalcemia as a condition, it will thus allow for a better diagnostic profile and allow for a well-planned and informed therapeutic strategy.

2.2 How Excessive Vitamin D Can Lead to Hypercalcemia

Often referred to as the "sunshine vitamin," vitamin D is vital to human health. It is essential for immune system performance, cell proliferation, and bone formation. Both food sources and solar exposure provide us with vitamin D. Nevertheless, too much vitamin D can have negative consequences, the most serious of which is hypercalcemia disorder marked by unusually elevated blood calcium levels. via a review of pertinent

scientific literature this article will examine three main ways via which hypercalcemia is caused by an overabundance vitamin D.

2.2.1 Intestinal Calcium Absorption on Overdrive

Encouraging the body to absorb calcium from the intestines is one of vitamin D's main roles. This is accomplished by vitamin D by inducing the production of proteins that carry calcium across the intestinal wall [12]. To make sure the body receives just the necessary quantity of calcium, this process is strictly controlled. However, this control system malfunctions when vitamin D levels get too high. "Chronically high vitamin D intake" causes a persistent increase in intestinal calcium absorption, which in turn causes an overflow of calcium to reach the circulation, as demonstrated by research published in the Journal of Clinical Investigation [12]. The result of this mismatch between calcium intake and excretion is hypercalcemia.

2.2.2 Mobilizing Calcium from the Bones: A Double-Edged Sword

Additionally, vitamin D affects bone tissue, affecting the way calcium is released. Under normal conditions, vitamin D encourages the absorption of calcium into the bone matrix, which aids in the maintenance of strong bones. For the growth and strength of bones, this process is essential. There is, nonetheless, a fine balance. Excessive vitamin D can cause an excessive release of calcium from bones into the circulation, according to research published in bone reports [13]. As a result, the body's equilibrium is upset, which can eventually weaken bones and cause hypercalcemia.

2.2.3 Suppressing the Body's Natural Calcium Regulator

By generating parathyroid hormones, the parathyroid glands are essential for preserving the calcium balance (PTH). PTH rises in response to a decrease in blood calcium levels, telling the body to release calcium from bones or boost intestine absorption. On the other hand, studies that were published in the Endocrine Journal indicate that too much vitamin D may inhibit the formation of PTH [14]. The body's natural feedback loop for controlling calcium levels is thrown off balance by this. In the absence of PTH's inhibitory action, calcium levels may increase uncontrollably and exacerbate hypercalcemia.

Hypercalcemia can result from an overabundance of vitamin D through a number of different processes. It reduces the body's natural PTH-mediated calcium control, increases intestinal calcium absorption, and upsets the delicate balance of calcium release

2.3 Mechanisms of Vitamin D-mediated Hypercalcemia

In a normal functioning body, vitamin D maintains calcium homeostasis as its job is to promote the absorption of calcium. Once vitamin D is produced by the body (in the form of cholecalciferol) or ingested by a human, it will enter the blood circulation and undergo the chemical reaction of hydroxylation in the liver followed by the same process in the kidney to make an active form of vitamin D known as 1,25-dihydroxyvitamin D or in other words calcitriol [15]. The active form of vitamin D binds to intestine receptors and result in the production of proteins whose jobs are to ease the reabsorption of calcium from food, these proteins are transport proteins known as TRPV6, Calbindin 9K and CX[16]. Excessive levels of the active form of vitamin D as shown in figure 1 [17], calcitriol, leads to an increased calcium reabsorption from the intestines through the transporter proteins mentioned above. It is a direct result of the cascade that occurs when calcitriol binds to the vitamin D receptors in the epithelial cells, because calcitriol causes the gene for the expression of the protein transient receptor potential vanilloid 5 to express more of it, these proteins act as channels for the reabsorption of calcium right back into the bloodstream. Not to mention that calcitriol also results in the production of calbindin-D, which simply makes the reabsorption of calcium more efficient [18].

The reabsorption of calcium from the intestine is a complex process that has many mechanisms and regulations, there are two pathways for calcium reabsorption, the paracellular pathway and the Transcellular pathway. calcium takes the paracellular pathway when there is more dietary calcium present, tight junction proteins control how much calcium goes through. Meanwhile the transcellular pathway, is more controlled, where proteins like TRPV6 work to either prevent or allow calcium entry, they are simulated by vitamin D. Proteins like calbindin bind to calcium inside the cell, and carry it to leave through another protein, a pump known as PMCA1b that pumps protein into the blood [19].

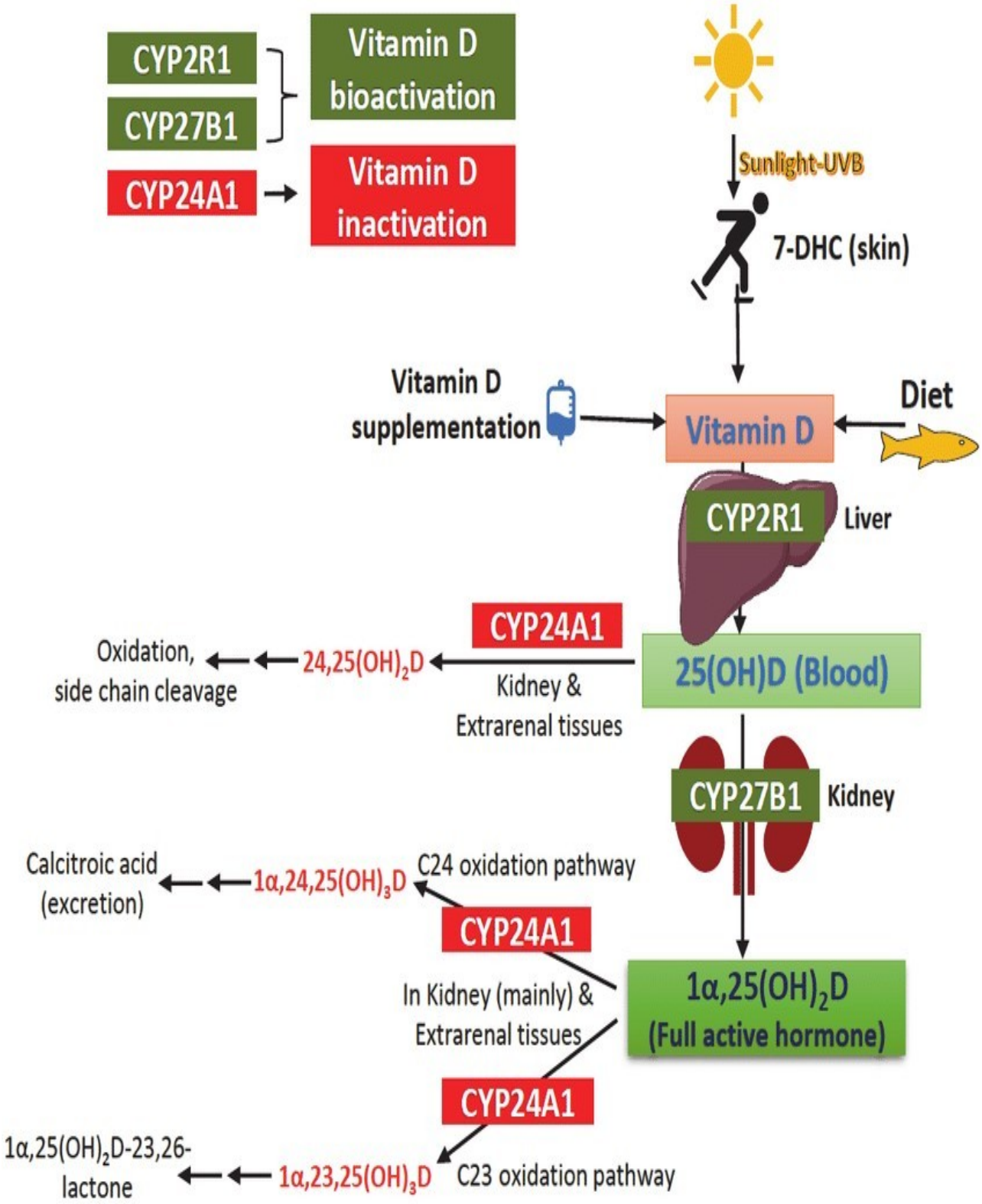

Figure 1 : Pathways of Vitamin D synthesis, we can see the result of 1,25dihydroxyvitamin D also known as Calcitriol

2.4 Primary Hyperparathyroidism and Vitamin D Levels

An increased reabsorption can in some cases result in too much calcium in the bloodstream, meanwhile, we now understand that excessive calcium can also simulate PTH production the bone which could lead to increased bone resorption and further increase the levels of calcium in the blood. This works through a negative feedback mechanism were normally if there is a healthy level of calcium in the blood, this would signal to the parathyroid glands to decrease PTH production which in turn decreases calcium in the blood stream, then it follows that once the calcium levels go below the normal threshold, it would also conversely simulate the parathyroid glands to release more PTH and in turn for more calcium in the blood.

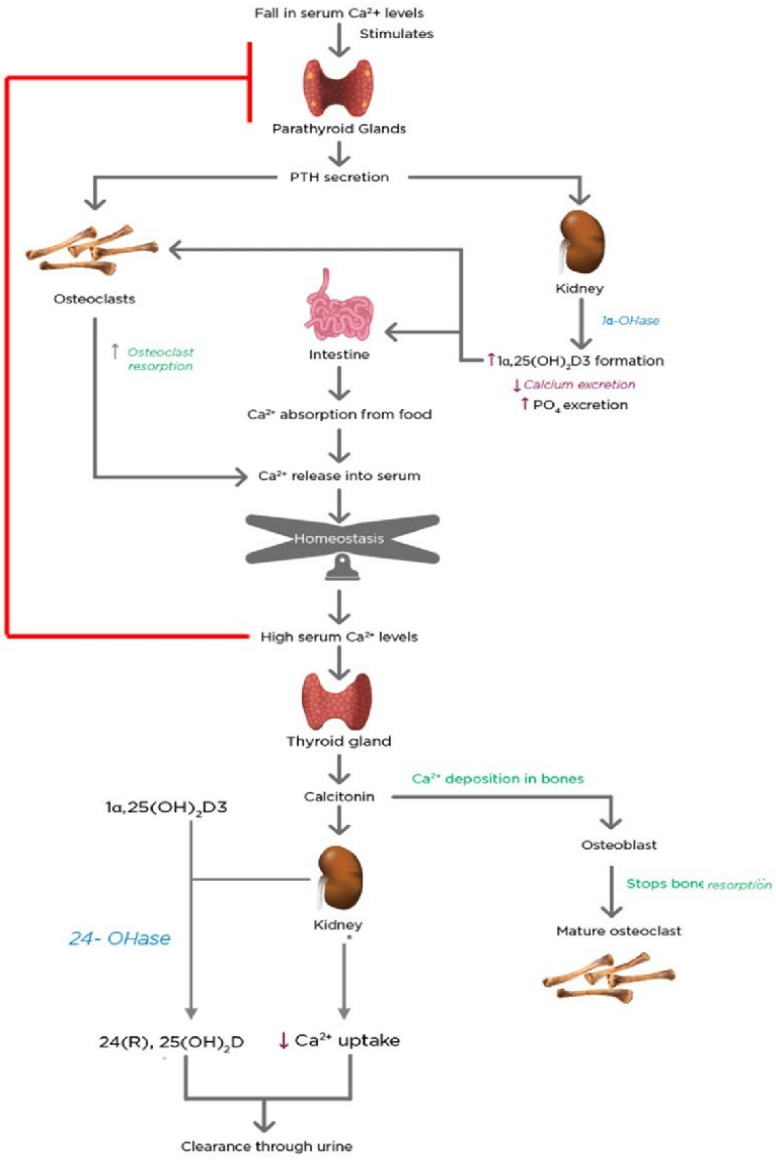

Figure 2 : shows the Negative Feedback Loop of PTH, Vitamin D and Calcium

Figure 2 [20].shows that vitamin D is a major steroid hormone that is gaining attention as a therapeutic molecule. Due to the general awareness of its importance for overall well-being, vitamin D deficiency (VDD) is now recognized as a major health issue. The main reason for VDD is minimal exposure to sunlight,

When hypercalcemia occurs because of vitamin D mediation, this negative feedback loop is disrupted, but this is an indirect process, as we alluded to earlier. In the human body, bones are remodeled continuously and for it to occur properly, the body uses to of its main cell types, the osteoblasts and the osteoclasts [21]. Osteoblasts are types of cells responsible for building bones and osteoclasts are cell types that work on bone resorption. As can be seen in the figure above, PTH plays a crucial role to maintain the calcium by focusing on the regulation of osteoclasts and osteoblasts, hence why it is considered an indirect mechanism. If calcium levels decrease below normal, the PTH simulates osteoblasts into producing a protein called RANKL which then goes on to bind to its receptors that are sticking out on the surface of osteoclasts, which in turn activates these cells to break down the bone. This is how bone resorption happens [22]. As such, bone resorption causes the release of calcium into the bloodstream raising calcium, until it reaches a level above the accepted threshold and PTH stops being produced. This mechanism is disrupted, in hypercalcemia, there is excess calcium and excess calcitriol, which disrupts the negative feedback loop, particularly, calcitriol can influence osteoblasts, by interfering with the normal suppression of PTH when calcium reaches that upper cap in the bloodstream, the feedback loop stops working if it cannot indicate to the parathyroid glands that PTH secretion needs to stop, so PTH continues to be secreted and they continue to cause osteoblasts to produce RANKL, which activates osteoclasts which breaks down more born, resorbing it and releasing even more calcium, as such hypercalcemia results [23]. This increase in PTH production is what characterizes the disorder known as primary hyperparathyroidism, which can have many causes, including a benign tumor known as parathyroid adenoma [24] which causes the excessive release of PTH, this is another cause outside of excessive calcitriol presence. In some other cases, hyperparathyroidism can be caused by hyperplasia of more than one of the four parathyroid glands [25] or even parathyroid cancer, this cancer is sometimes even diagnosed due to hypercalcemia in patients [26].

2.5 Granulomatous Diseases and Their impact on Vitamin D Metabolism

There are many reasons for why vitamin D elevations and excessiveness of it can cause hypercalcemia, particularly the excessive intake of vitamin D supplements [27]. However, there are many other causes for it. It is essential for medical professionals to understand the varying and complex etiological factors that also contribute to hypercalcemia as a condition, it will thus allow for a better diagnostic profile and allow for a well-planned and informed therapeutic strategy. A common way in which hypercalcemia is induced is when individuals with granulomatous diseases have elevated levels of calcium. Such diseases include sarcoidosis [28] or tuberculosis [29], in this case, the high level of calcium is easily attributable to the ability of immune cells to produce vitamin D in an active from, this happens because the immune cells themselves are activated and this is unique to such cells. This unique process if often referred do as ectopic production [30], it is able to overhaul the normal mechanisms of regulation set in place by homeostasis, particularly those induced and instructed by the kidneys, allowing for an elevated and abundant level of active vitamin D in the body. Granulomatous disorders: positive disorders, which include sarcoidosis or tuberculosis, are called granulomatous sicknesses. those situations can result in an unusual production of the lively shape of vitamin D through the immune cells gift within the granulomas. This heightened production increases the absorption of calcium from the gut, main to multiplied blood calcium levels. The link between those diseases and hypercalcemia is critical for healthcare vendors to apprehend, as it could affect each the prognosis and treatment of affected patients [31].

2.6 Malignancy-Associated Hypercalcemia Mediated by Vitamin D

HHM, which stands for Humoral Hypercalcemia of Malignancy is also a unique in the way it handles the metabolism of calcium. One of the things that makes it unique is the fact that it does not get triggered by excessive vitamin D, but rather it is a result of some cancers, especially lymphomas and some other solid tumors, become capable of producing paraneoplastic factors [32]. These factors are substances that are both chemically and physically similar to active vitamin D, however it is worth mentioning that it isn’t identical. This phenomenon, known as humoral hypercalcemia of malignancy, is a paraneoplastic syndrome, wherein the cancer cells secrete hormones or hormone-like

materials [33], inflicting systemic effects that are not related to the neighborhood presence of the cancer cells. Their impact and influence are on two key organs, intestines and bones, wherein they stimulate one to reabsorb Calcium [34] and promote resorption [35] in the other respectively, as a result they increase calcium levels in the bloodstream from figure 3 [36].

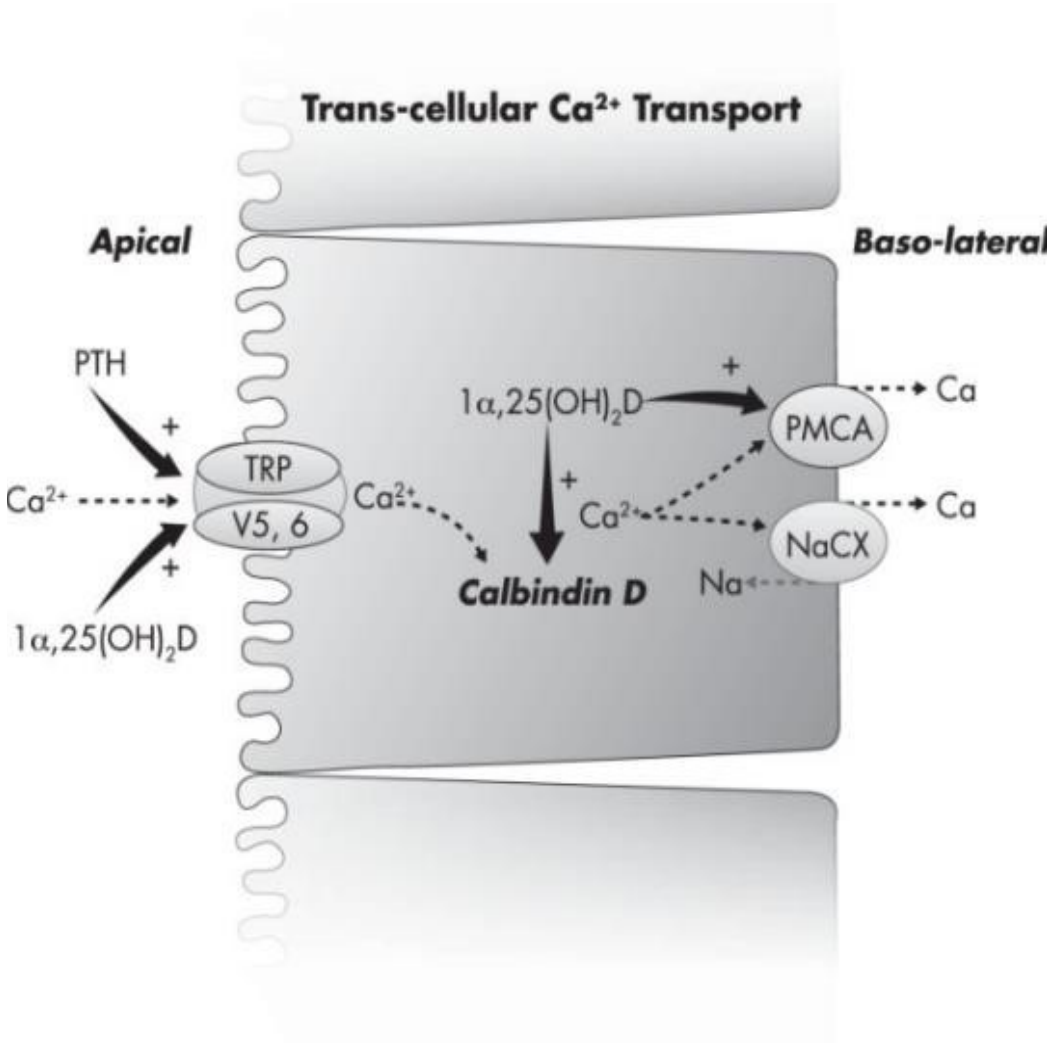

Figure 3 : Integrated model of activeCa2+ reabsorption in the intestine and distal part of the nephron

Chapter Three

Results and discussion

3.1 Clinical Manifestations and Laboratory Findings of Hypercalcemia:

As mentioned earlier, disruptions to the precariously sensitive balance and interplay of vitamin D and calcium, can result in hypercalcemia. It is crucial that the medical and scientific community as a whole understand the spectrum of potential symptoms and manifestations of the medical condition. Naturally, as is the case with many complicated ailments or conditions, these symptoms can vary in the severity, in the level of debilitation and in how it affects the patient's fitness levels. The severity levels are proportional to the duration of the calcium elevation but also to the degree of elevation of calcium in the bloodstream. Not to mention, the systems that hypercalcemia affects can vary.

As such, hypercalcemia can have an immense and significant effect on a patient's nervous system. For example, the patient may, at the start, initially experience some subtle symptomatic manifestations, ranging from fatigue to lethargy. However, when the calcium levels rise further, there will be severe manifestations, patients will experience symptoms like confusion, disorientation or impairments in their cognitive abilities. These neurological modifications are the idea to result from the disruptive consequences of high calcium stages on neuronal features and the way in which the patient's brain does metabolism. There are actual research done on how it can affect the neurological system, at focal points [37].

Beyond the neurological system, hypercalcemia also affects a patient's gastrointestinal system. The signs and symptoms of course, varies according to range, elevation and time.

Excessive calcium ranges can affect the gastrointestinal system significantly, a patient may experience symptoms such as nausea, vomiting, constipation, and abdominal pain. Those symptoms arise due to the fact calcium plays a function in the contraction of easy muscle, and its excess can disrupt the ordinary motility and function of the gastrointestinal tract [38].

There are also some cardiac implications of hypercalcemia, naturally, because calcium is an ion that is integral to muscle contractions, including the contraction of heart muscles, one of the critical areas tormented by hypercalcemia is the cardiovascular system itself. calcium is essential to the regulation of coronary heart muscle's features and the electric conduction device of the heart [39], that is, the sinoatrial node and the atrioventricular node. Consequentially, an elevated level of calcium, i.e., hypercalcemia can lead to cardiac arrhythmias [40], which could range from a benign arrythmia to a more serious and life-threatening rhythm. At the same time, that very same elevated calcium levels can contribute to excessiveness in high blood pressure levels and can be attributed to an elevated danger of coronary heart attack.

A final affected or targeted system within the human body worth looking at is the musculoskeletal system. Again, this is largely attributed to the role of calcium in muscle contractions. There are a range of signs and symptoms: within the context of vitamin D mediated hypercalcemia, bone fitness can be adversely affected. It is widely known that vitamin D is critical for bone health, its excess, mainly while blended with elevated calcium levels, can lead to bone tenderness and pain as a result of it. This happens due to the elevated bone turnover and ability bone resorption, as the body tries to normalize calcium elevations [41].

It is essential to recognize these symptoms of hypercalcemia in order to have effective treatment plans in place and alleviate the possible health complications that will result. Of these complications which may be shown in clinical and lab findings are: bone pain and fractures as a result of bone resorption in hyperparathyroidism patients with high levels of PTH [42], kidney stones which may result from the high calcium levels that manifests in the urine and even block urine flow [43], this condition is considered rare but is still important to note. Finally, as mentioned earlier in this section, calcium ion is incredibly important in cardiac function and ad regulating heart beats, having an excess of it may show up as cardiac arrythmia [44].

3.2 Diagnostic Algorithms for Identifying Vitamin D-Mediated Hypercalcemia

The manifestations of the symptoms of hypercalcemia are hard to distinguish on its own, for example, arrythmia could be the result of any number of diseases that are not necessarily related to hypercalcemia, not to mention, the underlying causes of hypercalcemia are also difficult to ascertain. We found out earlier for example that parathyroid cancer is one of the rare underlying causes of hypercalcemia in patients. What then is the specific algorithm to be followed when attempting to diagnose hypercalcemia and its underlying causes.

For starters, an algorithm is a series of sequential steps to be followed, that means it includes the standard taking of patient history, doing the physical examination and so on. At these first two stages, a diagnostician may develop some suspicions, for example, the patient may have muscle weakness, dehydration and abdominal tenderness, all of which are symptoms of hypercalcemia [45], the patient may even present with nausea or vomiting, to some extent kidney stones too, the reasons for which are highlighted in the earlier section. Other tools can be bone density measurements as we know that PTH levels can mean more breakdown of bones that has resulted in a high calcium level.

The standard for testing for hypercalcemia is to check the serum calcium levels of the patient, it is one of the most primary ways that hypercalcemia is diagnosed, this can be followed by checking the vitamin D levels to discern if it is indeed mediated vitamin D or is it a perithyroidal issue. As such, the diagnostician would also be wise to take a PTH level measurement and finally to check for some other electrolytes as it is important to assess kidney function considering its importance in both making active vitamin D and calcium absorption as demonstrated.

The use of imaging in diagnosis was also noteworthy, with the most common ones used imaging techniques being parathyroid scintigraphy, although it seems in the US they reserve CT and MRI scans for patients with recurring hypercalcemia [46].

Hypercalcemia is now considered to be at a level greater than the threshold the normal range of (8.9 mg/dL - 10.4 mg/dL), this can be measured in the serum calcium levels test which measures the total calcium level in the body [47], meanwhile a PTH level of greater than 25 pg/mL indicates abnormality [48], however, this isn’t always a

good indicator as non-parathyroid causes of hypercalcemia can have low levels of PTH, it is why the serum calcium level is a standard test.

3.3 Treatment Options for Hypercalcemia

The first line treatment for hypercalcemia is intravenous fluids, especially if the case is severe, and the degree to which it is effective is directly from how it can respond to two aspects of the hypercalcemia condition, the first being the ability to dilute, as it essentially dilutes blood plasma when it increases the volume and as a result it will decrease the concentration of calcium and it will immediately alleviate some of the extreme symptoms. At the same time, it enhances excretion, because an increased level of fluid also results in diuresis allowing for quick excretion of calcium in the body. In recent papers, it was highlighted that while isotonic saline has traditionally been used, it might be better to use isotonic crystalloid solutions as it helps to minimize the electrolyte imbalance that will result [49]. There are also some suggestions that if medical professionals start the therapy early and target the volume based on the individual patients, there will be better outcomes [50]. This theme of individualized patient care is one of the key results of this literature review.

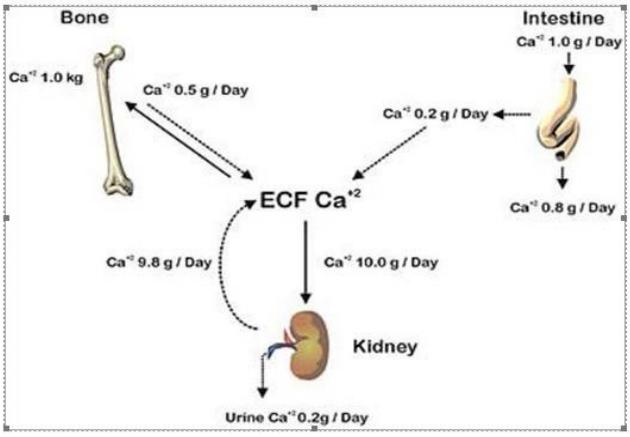

Figure 4 : Calcium balance. On average, in a typical adult approximately 1g of elemental calcium (Ca) is ingested per day

Another therapeutic method is through the promotion of calcium excretion by using loop diuretics as shown in figure 4 [51], there are the benefits of the quick action in treatment especially if you combine. this treatment with the IV fluid therapeutic solution. It has the additional benefit of efficient hydration if used in tandem with it, allowing dehydration, simulate diuretics and increase calcium excretion. However, there is that risk of dehydration, so there needs to be a monitoring of electrolyte levels while the treatment

is being done, as it can also very easily cause electrolyte imbalances. Not to mention the body can develop resistance to this treatment however, as usual, the latest research indicates an individualized approach in the treatment, because as it might be expected, the effectiveness of the treatment is affected by the severity of hypercalcemia in the individual patient, and an ideal dose is recommended per individual patient [52]. As alluded to earlier, this treatment can also be incredible efficient if combined with the IV fluid treatment, however some newer researchers are also suggesting using it with medications of bisphosphonates and calcitonin [53].

The way they function is as an inhibitor to osteoclast activity, which are actually cells that function to break down bone tissue [54]. In this way it can not only lower calcium levels but actively prevent its release back into the bloodstream [55]. It has a targeted action and is more versatile and long-lasing than previously mentioned treatments. However, there are some potential side effects for example it can cause nausea, vomiting or abdominal pain and even in some rare cases lead to osteonecrosis in the jaw [56].

contrastingly, calcitonin is a natural hormone that is only a temporary measure when treating hypercalcemia, it is produced by the thyroid gland and it primarily works on inhibiting osteoclasts and as such preventing the release of calcium from bone breakdown, it also works on an increase of the excretion of calcium via the urinary tract [57]. That is, it functions similarly to a diuretic, a mild one at that, to promote it. It is often used for patients who have a resistance to the treatments of bisphosphonate [58]. It is worth mentioning it is less effective than loop diuretics however, this may also mean milder symptoms. It is often the case that loop diuretics are used in combination with calcitonin [59]. Unlike bisphosphonate, calcitonin is short lived with limited efficiency, while also being liable to cause the body to build a tolerance towards it.

3.4 Discussion

This review has delved into the intricate interplay between vitamin D, calcium, and the development of hypercalcemia [60]. While recent studies have shed light on the potential dangers of excessively high vitamin D levels [61], a more comprehensive understanding is necessary. Vitamin D acts as a key player in calcium homeostasis, fostering bone health and regulating calcium absorption within the intestines [62]. However, as demonstrated in numerous studies [63] exceeding recommended vitamin D intake or disruptions in the

body's regulatory mechanisms can lead to hypercalcemia, a condition characterized by an abnormal elevation of blood calcium levels.

This review emphasizes the critical importance of tailoring treatment approaches for hypercalcemia caused by vitamin D. Research has shown the effectiveness of intravenous fluids, loop diuretics, bisphosphonates, and calcitonin as potential therapeutic options. However, the effectiveness and potential side effects of these treatments vary significantly depending on the severity of the condition, the underlying cause, and the patient's unique characteristics. As reported by Mithal [64] and Kaufmann et treatment strategies must be meticulously crafted for each individual patient. A comprehensive diagnostic approach is crucial for determining the most suitable treatment plan. This approach goes beyond simply measuring serum calcium levels and should encompass vitamin D levels, PTH levels, and other relevant factors such as kidney function and any co-existing medical conditions. As advocated by Bouillon et al. (2019) [65], a more holistic diagnostic evaluation allows for a targeted and effective treatment plan that minimizes the risk of complications while maximizing patient benefit.

Hypercalcemia caused by elevated vitamin D levels highlights the delicate balance between promoting bone health and maintaining calcium homeostasis. A deeper understanding of the underlying mechanisms and a shift towards more individualized treatment approaches, as evidenced by the reviewed studies, are essential for ensuring optimal patient care. Furthermore, ongoing research emphasizes the importance of individualized treatment approaches for vitamin D-induced hypercalcemia. Like a chessboard where each piece has its unique role, various therapeutic options like intravenous fluids, loop diuretics, bisphosphonates, and calcitonin offer potential solutions [66]. However, the effectiveness and potential side effects of these therapies vary significantly depending on the severity of hypercalcemia, the underlying cause, and the patient's individual characteristics. A comprehensive diagnostic approach is crucial, considering not only serum calcium levels but also vitamin D levels, PTH levels, and other relevant factors. This multifaceted approach, as advocated by researchers allows for a more targeted and effective management plan, minimizing complications and maximizing patient benefit [67].

Chapter four

Conclusion & Recommendations

4.1 Conclusion

The review provides an in-depth exploration of hypercalcemia, focusing on its relationship with vitamin D and its implications for health. It highlights the need for a nuanced approach considering both risks and benefits. The objective is to critically analyze existing literature and advocate for a more nuanced therapeutic strategy.

In conducting this review, we saw that there is a complexity behind the symptoms, the causes, the treatment and understanding of the medical condition of hypercalcemia. It was important to the researchers of this paper to highlight that an umbrella assumption that over consumption of vitamin D causes hypercalcemia is too simplistic and broad to result in any long--term effective solution to the long-term treatment of hypercalcemia.

Most important finding of this review was the emphasis on individualized treatment of patients of hypercalcemia, as we found out that there are specific and underlying causes for it. The paper also demonstrates that different concentrations of calcium results in different symptoms in different patients, and it is only through understanding all the different ways in which hypercalcemia can we result do we understand how to alleviate the symptoms and restore homeostasis in individual patients.

4.2 Recommendation

1. Individualized treatment is key to treating hypercalcemia as there are varied etiologies and symptoms that are diverse from patient to patient. This is now considered one of the optimal ways in which this condition is treated internationally.
2. Conduct comprehensive studies to explore lesser-known etiological factors, optimize treatment modalities, identify predictive biomarkers, and assess long-term outcomes of hypercalcemia, facilitating personalized and effective management strategies
3. Early diagnosis and intervention is paramount, several of the papers highlight the importance of early diagnosis of hypercalcemia and its underlying causes for effective treatment, it is suggested that both calcium and vitamin D monitoring at regular intervals is one of the best preventative measures and allows for early diagnosis.

4. Patient education and adherence is also useful when treating hypercalcemia, with a comprehensive understanding of how the underlying mechanism and interplay of vitamin D and calcium work in tandem to each other we could hope to prevent any excessive supplementation of either.
5. Preventative measures should be highlighted amongst healthcare professionals as it seems to play a key role in preventing an imbalance in the first place, patients should be kept within the recommended levels of vitamin D and regularly be asked to check up, particularly high-risk patients, such as those suffering from renal failure and heart disease.

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